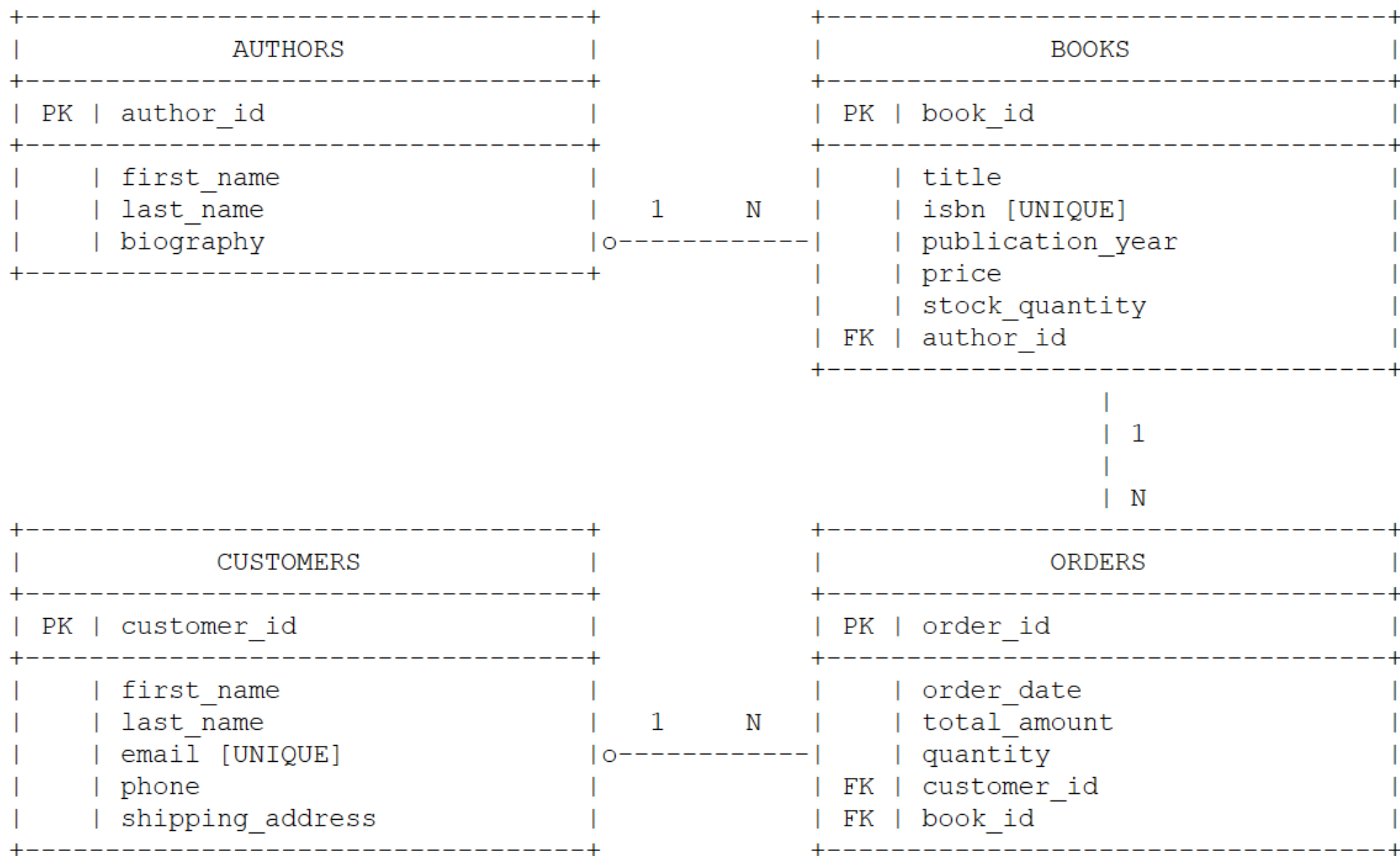


WEEK 5 - EXERCISE

MONDAY

ERD



AUTHORS TABLE (Independent Entity)

- **Attributes:**
 - author_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each author.*
 - first_name (Text, NOT NULL, CHECK) → *Cannot be null or empty spaces.*
 - last_name (Text, NOT NULL, CHECK) → *Cannot be null or empty spaces.*
 - biography (Text, Nullable).
- **Data Integrity Constraints:**
 - CHECK(length(trim(first_name)) > 0)
 - CHECK(length(trim(last_name)) > 0)
- **Relationship:** Connects to **books** with a **One-to-Many (1:N)** line.

BOOKS TABLE (Dependent on authors)

- **Attributes:**

book_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each book.*

title (Text, NOT NULL, CHECK) → *Cannot be empty.*

isbn (Text, UNIQUE, NOT NULL, CHECK) → *Must be distinct; prevents duplicate listings of the same edition.*

publication_year (Integer, CHECK) → *Must fit realistic chronological boundaries.*

price (Real, NOT NULL, CHECK) → *Financial value must be positive.*

stock_quantity (Integer, NOT NULL, DEFAULT 0, CHECK) → *Defaults to zero; cannot be negative.*

author_id **[FK]** (Integer, NOT NULL) → *References authors.author_id.*

- **Data Integrity Constraints:**

CHECK(length(trim(title)) > 0)

CHECK(length(trim(isbn)) >= 10)

CHECK(publication_year >= 1455 AND publication_year <= strftime('%Y', 'now'))

CHECK(price > 0.0)

CHECK(stock_quantity >= 0)

- **Referential Integrity Rule:**

ON DELETE RESTRICT → *Prevents deleting an author if they still have books linked to their record.*

- **Relationship:** Connects to **orders** with a **One-to-Many (1:N)** line.

CUSTOMERS TABLE (Independent Entity)

- **Attributes:**
 - customer_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each customer.*
 - first_name (Text, NOT NULL, CHECK) → *Cannot be empty.*
 - last_name (Text, NOT NULL, CHECK) → *Cannot be empty.*
 - email (Text, UNIQUE, NOT NULL, CHECK) → *Must be distinct for logins and follow standard email patterns.*
 - phone (Text, Nullable, CHECK) → *Optional, but must meet basic length requirements if provided.*
 - shipping_address (Text, NOT NULL, CHECK) → *Required for shipping calculations.*
- **Data Integrity Constraints:**
 - CHECK(length(trim(first_name)) > 0)
 - CHECK(length(trim(last_name)) > 0)
 - CHECK(email LIKE '%_@__%._%') → *Validates basic format string structure (e.g., user@domain.com).*
 - CHECK(phone IS NULL OR length(phone) >= 7)
 - CHECK(length(trim(shipping_address)) > 0)
- **Relationship:** Connects to **orders** with a **One-to-Many (1:N)** line.

ORDERS TABLE (Dependent transaction entity)

- **Attributes:**

- order_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each transaction.*
- order_date (Text/Timestamp, NOT NULL, DEFAULT CURRENT_TIMESTAMP) → *Automatically timestamps the record upon generation.*
- total_amount (Real, NOT NULL, CHECK) → *Total financial balance must be non-negative.*
- quantity (Integer, NOT NULL, DEFAULT 1, CHECK) → *Must purchase at least 1 book per transaction.*
- customer_id **[FK]** (Integer, NOT NULL) → *References customers.customer_id.*
- book_id **[FK]** (Integer, NOT NULL) → *References books.book_id.*

- **Data Integrity Constraints:**

- CHECK(total_amount >= 0.0)
- CHECK(quantity > 0)

- **Referential Integrity Rules:**

- FOREIGN KEY (customer_id) REFERENCES customers(customer_id) ON DELETE RESTRICT
- FOREIGN KEY (book_id) REFERENCES books(book_id) ON DELETE RESTRICT

Note: *Both rules guarantee that transactional sales history cannot be broken or orphaned by deleting an active customer or a cataloged book.*

EMPLOYEES TABLE

Column	Data Type	Constraints	Purpose
employee_ID	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique identifier for each staff member.
first_name	TEXT	NOT NULL, CHECK(length(trim(first_name)) > 0)	Prevents empty/blank names.
last_name	TEXT	NOT NULL, CHECK(length(trim(last_name)) > 0)	Prevents empty/blank names.
department	TEXT	NOT NULL, CHECK(department IN ('Marketing', 'Sales', 'Customer Support', 'Engineering'))	Restricts values to valid company departments only.
hired_date	TEXT	NOT NULL	Uses YYYY-MM-DD format (SQLite default).

PRODUCTS TABLE

Column	Data Type	Constraints	Purpose
product_ID	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique identifier for each product.
product_name	TEXT	NOT NULL, CHECK(length(trim(product_name)) > 0)	Descriptive name.
category	TEXT	NOT NULL	Categorizes the item (e.g., Electronics).
price	REAL	NOT NULL, CHECK(price > 0)	Ensures the price is always a positive number.
units_in_stock	INTEGER	NOT NULL, DEFAULT 0, CHECK(units_in_stock >= 0)	Starts at 0; prevents negative inventory.
units_sold	INTEGER	NOT NULL, DEFAULT 0, CHECK(units_sold >= 0)	Tracks total sales count.
discontinued_date	TEXT	NULLABLE	Optional; only populated if the item is removed from catalog.

RECOMMENDED SCHEMA (Spotify)

Column	Data Type	Constraints	Reasoning
id	TEXT	PRIMARY KEY	Your data uses unique alphanumeric codes for the ID, so this is the perfect Primary Key.
track_name	TEXT	NOT NULL	The core identifier of the entry.
artist	TEXT	NOT NULL	Standardizing this as text.
album	TEXT	NOT NULL	Standardizing this as text.
release_date	TEXT	NOT NULL	SQLite uses TEXT for dates. Using YYYY-MM-DD format allows for easy sorting.
popularity	INTEGER	CHECK(popularity >= 0 AND popularity <= 100)	Popularity metrics usually exist on a scale of 0–100. This constraint prevents illogical values (like -5 or 999).
duration_min	REAL	CHECK(duration_min > 0)	Prevents negative or zero-length tracks.
spotify_url	TEXT	NOT NULL	Ensures every track record has a valid link.